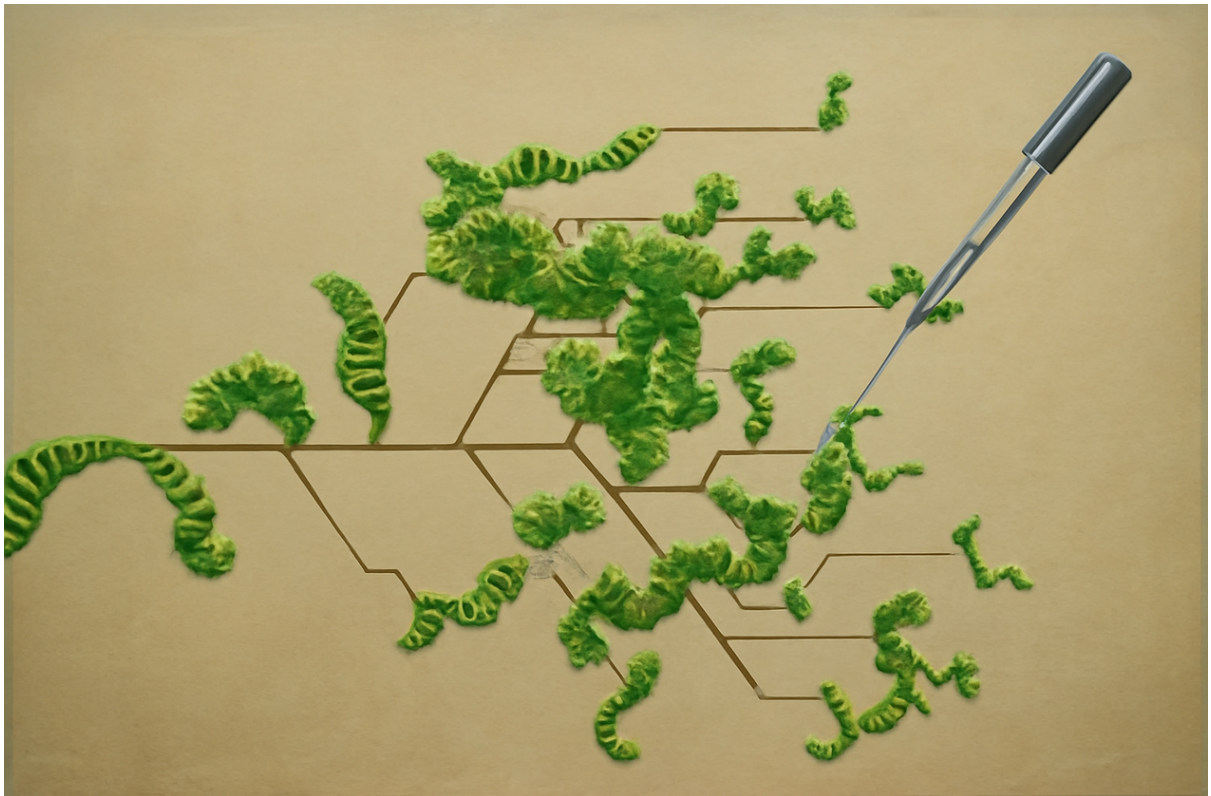


Short and long time-scales in Speciation Genomics: Machine Learning approaches



University of
St Andrews

UNIVERSITÄT **BONN**



Biotechnology and Biological Sciences Research Council

This meeting was funded in part by the UKRI Biotechnology and Biological Sciences Research Council (BBSRC) and the University of St Andrews and University of Bonn Collaborative Research Grant.

Faculty: Nick Bailey, Rui Borges, Carolin Kosiol, Antonio Pacheco, Mike Ritchie (University of St Andrews), Svitlana Braichenko (University of Edinburgh), Martin Kapun (Natural History Museum of Vienna), Manolo Perez (Imperial College London), and Alexander Suh (Research Museum Koenig Bonn)

Workshop Demonstrators: Manel Ait El Hadj, Omar Ruelas, Diogo Ribero, Lera Telizhenko (University of St Andrews), Ioanna Kotari (University of Vienna),

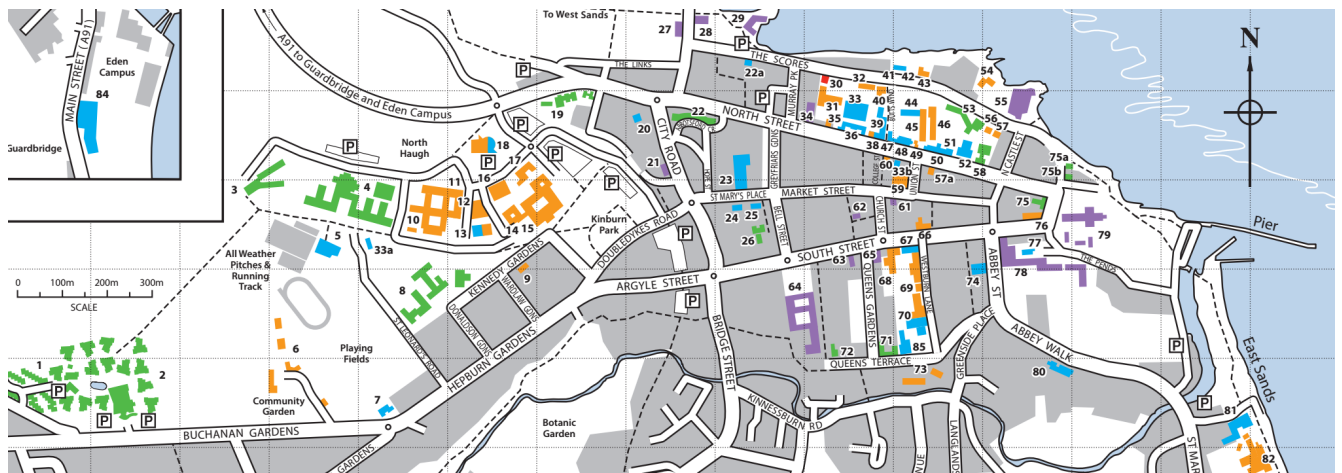
This abstract booklet was arranged by Nick Bailey.

St Andrews Stuff was written by Mike Ritchie, with very minor changes by Nick Bailey

Cover image “Extracting DNA from a Phylogeny” was generated with DeepAI and thus can be used for any legal purpose.

All presentations (workshops, lectures, posters, talks) are ordered alphabetically by the first author’s last name. Workshops and lectures are noted as such in the title.

Information on Locations



Map of St Andrews. Notable locations using numbers on map are:

Accommodation (4) at Agnes Blackadder Hall west of the town around North Haugh
 Lectures and Workshops (31) at Arts Building in the north middle of town on North Street
 Conference Dinner (44) at Lower College Hall in the same cluster of buildings as above
 Posters at Bell Pettigrew Museum (70) in the south middle of town below South Street
 St Andrews Bus Station (21) on City Road



Starting from top left and going clockwise the buildings are Agnes Blackadder Hall, the Arts Building, Lower College Hall, St Andrews Bus Station, and Bell Pettigrew Museum.

Recommendation for travel from abroad: From Edinburgh Airport the best option is the JET787 bus (<https://www.stagecoachbus.com/promos-and-offers/east-scotland/jet-747-edinburgh-airport-bus>). This bus arrives at St Andrews Bus Station so when you arrive you can mentally place a “YOU ARE HERE” sticker on Bus Station location in the above map.

Leisure activities in St Andrews can be found in “St Andrews stuff” section at the end of this booklet.

11. AV/IT

24 Jun to 27 Jun

IT EQUIPMENT – INTERNAL USERS

The standard classroom teaching equipment can be used by staff and students.

Operating System/Software used and supported within the University:

Windows 11
Microsoft Office 365
Web browser
PDF reader
Panopto (video capturing software for presentations/meetings)

General:

All presenters/speakers should be advised to bring a copy of their presentation on a USB drive. It is the responsibility of the Event Organiser & Speaker to check in advance that presentations will work correctly on the PC software that we provide. Anyone choosing to create a PowerPoint presentation on a Mac should be advised to convert their presentation to a Windows PowerPoint format [if not bringing their own Mac].

Presenters/Speakers who use their own laptop:

Should presenters choose to use their own laptop, it is their responsibility to make sure that any video or live internet links work correctly on site.

Apple Mac Users:

An Apple Mac Laptop is not available for loan within the University.

WIFI – Summer Time

Wifi can be accessed by your delegates via two methods:

1. Eduroam -
The University of St Andrews has eduroam - a global service that enables students, researchers and staff from participating institutions to obtain wifi connectivity. This is available in most University of St Andrews buildings (including residences). If your home institution has eduroam available, please ensure this is set-up prior to your visit to St Andrews. You will be unable to set up an eduroam connection in St Andrews if you haven't already set it up where you work or study.
2. BT Openzone is also available to guests without wifi. You will be able to connect free of charge in most University of St Andrews buildings (including residences).

YOUNG PERSONS

Please note the University does not and cannot filter its web content. It is your responsibility to manage the access to the internet for your students/attendees.

24 Jun to 28 Jun

IT EQUIPMENT – EXTERNAL USERS

The standard classroom teaching PC can be only used by staff and students of the University of St Andrews.

Presenters/Speakers need to bring their own laptop:

External organisers, presenters and speakers should use their own laptop. They can connect the laptop to our projection system. It is their responsibility to make sure that any video or live internet links work correctly on site.

The University is not able to supply a laptop.

Connectors:

- HDMI connection

****Apple Mac users need to bring their own adapter to be able to connect to the HDMI connector****

AV-Demo - Technician to meet client:

Date: Wednesday 25th June
Time: 0900
Location: Arts seminar 1 room 115

Specific AV requirements:

AV demo, attend and assist

12. Security & Response

Workshop ML Speciation Genomics 24th June - 27th June 2025

Arrival participants 24th June 2025

Event are in the Arts Building (Level 1 Foyer, Sem Room 115, 117 & 119)

Wednesday 25th June 2025

- 09:00-09:30 *Registration*
09:30-09:45 *Welcome* (Carolyn Kosiol & local St Andrews organizers)
09:45-10:30 **Lecture 1** Mike Ritchie (St Andrews): Does experimental evolution recapitulate speciation genomics?
10:30-11:00 Coffee Break
11:00-11:45 **Lecture 2** Alex Suh (Bonn): The contribution of genomic (in)stability to population and species divergence
11:45-12:30 **Contributed talks**
11:45-12:00 Francisco Ruiz-Ruano (Bonn): Evolutionary dynamics of GRCs in passerine birds
12:00-12:15 Diogo Ribeiro (St Andrews): Uncovering the architecture of selection from experimentally evolved populations with Machine Learning
12:15-12:30 Valeriia Telizhenko (Schmalhausen Institute of Zoology, Kyiv): A Machine Learning approach to identifying recent selection in blue whale evolution
12:30-13:15 Lunch
13:15-15:30 **Workshop 1** Martin Kapun (Natural History Museum Vienna): From Museums to Microbes: A concise introduction to Museomics using *Drosophila melanogaster* and its symbionts as a model
15:30-16:00 Coffee Break
16:00-16:45 **Lecture 3** Svitlana Braichenko (University of Edinburgh): Phylogenomics meets Deep Learning
17:00-19:00 *Poster Session Bell Pettigrew Museum*

Thursday 26 June 2025

- 09:00-09:45 **Lecture 4** Christoph Mayer / Peter Arsic (Bonn): Basics in Neural Networks
09:45-10:30 **Contributed talks**
09:45-10:00 İdil Çağla Ertaş (Bonn): The genetic bases of quantitative morphological variation in a population of *Drosophila*
10:00-10:15 Nina Casillas (Bonn): Genomic and Ecological Insights into Biodiversity Decline: A Case Study of the Eurasian curlew, *Numenius arquata*
10:15-10:30 Manel Ait El Hadj (St Andrews): Polymorphism-aware Framework to Analyse RNA-Seq Data for Trait Evolution Studies
10:30-11:00 Coffee Break
11:00-12:30 **Workshop 2** Rui Borges (St Andrews): Detecting Targets of Selection in Experimental Evolution
12:30-14:00 Lunch
14:00-15:30 **Workshop 3** Christoph Mayer / Peter Arsic (Bonn): Machine learning topics in sequence analyses
15:30-16:00 Coffee Break
16:00-16:45 **Lecture 5** Manolo Perez (Imperial College London/ Madrid): PhyloCNN
18:30 -21:00 *Conference Dinner at Lower College Hall*

Friday 27 June 2025

- 09:00-09:45 **Lecture 6** Carolyn Kosiol (St Andrews): Polymorphism-aware phylogenetic models (PoMos)
09:45-10:30 **Contributed talks**
09:45-10:00 Ioanna Kotari (Vienna): PoMo-cod: a polymorphism-aware phylogenetic codon model
10:00-10:15 Menno de Jong (Leipzig): Species delineation using genome-wide Dxy- and Fst-estimates
10:15-10:30 Yari Cerruti (St Andrews): Sequence composition and conservation predict the phenotypic relevance of transposable elements
10:30-11:00 Coffee Break
10:30-12:30 **Workshop 4** Antonio Pacheco, Nick Bailey (St Andrews): Phylogenomics DeepBlues Project: balancing selection
12:30-14:00 Lunch
14:00-15:30 **Workshop 5** Carolyn Kosiol (St Andrews): Phylogenomics: PoMoBalance and RevBayes
15:30-16:00 **Discussion (All)**
16:00 Coffee, End of conference, Hike along the beach for participants if wanted

Posters:

Elisabeth Karalashvili (Bonn): Genomic erosion footprint in insects under extreme decline and isolation
Nok Ting Lam (Oulu): Genetic structure, spectral characteristics and evolutionary history of butterflies in the European *Polyommatus eros* species group (Lepidoptera, Lycaenidae)
Roni Loudon (St Andrews): Phylogenetic history of ATP phosphoribosyl transferase (ATPPRT)
Omar Ruelas (St Andrews): Adding Morphological Traits to the Species Tree inference under a Polymorphism-Aware Phylogenetic Models approach
Some more? All participants are encouraged to bring a poster if they like.

Polymorphism-aware Framework to Analyse RNA-Seq Data for Trait Evolution Studies

Manel Ait El Hadj

Centre for Biological Diversity, University of St Andrews, St Andrews, United Kingdom

Phylogenetic inference is complicated by processes such as incomplete lineage sorting (ILS), where gene tree discordance arises from the stochastic retention of ancestral polymorphisms across speciation events. While integrating polymorphisms into phylogenetic models can improve resolution of ILS, current approaches—despite advances in DNA-based methods like the Polymorphism-aware phylogenetic Model (PoMo), remain underdeveloped for RNA data. Classical differential expression tools such as DESeq2 and edgeR are designed for single populations and single genes and they ignore the phylogenetic context. To address this gap, we present a computational framework integrating RNA-Seq simulations, phylogenetic comparative methods, and machine learning to model polymorphisms and trait evolution. Our approach begins with *in silico* RNA-Seq data generation using the `compcodeR` package (Bioconductor), which simulates gene expression matrices under Brownian motion (BM) or Ornstein-Uhlenbeck (OU) models, conditioned on preset phylogenies and evolutionary scenarios. Simulated datasets are then combined with trait evolution analyses in `RevBayes` to infer parameters under BM/OU processes. Machine learning models trained on simulated data, are then applied to empirical RNA-Seq datasets to detect subtle evolutionary patterns. We apply this approach using Mexican Goodeinae, a clade of viviparous fish exhibiting divergent reproductive strategies and sexual selection dynamics. By simulating RNA-Seq data reflective of ancestral polymorphisms, we reconstruct trait evolution while accounting for genetic variation typically excluded in traditional phylogenies. Our approach advances scalable methods for disentangling the roles of selection, standing genetic variation, and speciation drivers in adaptive diversification.

Workshop: Detecting Targets of Selection in Experimental Evolution

Rui Borges

School of Mathematics and Statistics, University of St Andrews, St Andrews, United Kingdom

Experimental evolution allows us to observe evolution in real time as populations adapt to specific stressors, such as changes in temperature or diet. This approach is especially useful for identifying causal genetic variants that drive adaptation - for example, variants that increase in frequency during the experiment. In this hands-on workshop session, we will analyze time-series frequency data from experimental evolution studies. We will compare two methods for detecting targets of selection: a mechanistic approach ([BAIT-ER](#)) and a deep learning-based method. Finally, we will discuss the strengths and limitations of each approach and explore the biological significance of the results.

Lecture: Phylogenomics meets Deep Learning

Svitlana Braichenko

Institute of Genetics and Cancer, University of Edinburgh, Edinburgh, United Kingdom

Deep learning (DL) has been widely used in various scientific fields, but its integration into phylogenetics has been slower, primarily due to the complex nature of phylogenetic data. The studies that apply DL to sequencing data often limit analyses to 4-taxon trees. Many of these studies serve as “proof of principle” and perform similarly to traditional phylogeny reconstruction methods. New ways of using training data, such as encoding with compact bijective ladderized vectors or transformers, enable the handling of much larger trees and genomic data sets. This review focuses on the application of DL in phylogenetics, introducing prevalent DL architectures. We highlight potential problems in the field by discussing the risks of using simulation-based training data and emphasize the importance of reproducibility and robustness in computational estimates. Finally, we explore promising research areas, including the combination of phylogenetics and population genetics in DL, the analysis of neighbor dependencies, and the potential to significantly reduce computational cost compared to traditional methods. This perspective illustrates the potential of DL in complementing traditional phylogeny reconstruction methods and aiding the advancement of phylogenetic analysis, especially in performing computationally demanding tasks such as model selection or estimating branch support values.

Genomic and Ecological Insights into Biodiversity Decline: A Case Study of the Eurasian curlew, *Numenius arquata*

Nina Casillas¹, Eckart Stolle¹, Astrid Böhne¹, Verena Rupprecht³, Till Töpfer¹, Marianna Simões²

¹Leibniz Institute for the Analysis of Biodiversity Change, Museum Koenig Bonn, Bonn, Germany

²Senckenberg Research Institute and Natural History Museum, Frankfurt, Germany

³LBV - Landesbund für Vogel- und Naturschutz

The rapid loss of global biodiversity is an urgent issue that threatens all life on Earth. Anthropogenic activities leave fragmented habitats lacking heterogeneity in landscape causing a decline in species richness. With a growing number of species at risk—and many others not regularly assessed—there is a critical need to re-evaluate how extinction risk is measured. Traditional population monitoring alone is no longer sufficient; genomic analysis must be integrated into conservation assessments. This need has inspired the creation of the BiGFoot project (Biodiversity Decline's Genomic Footprint), which investigates population decline in threatened species using genomic data based on collection material. By analyzing historical DNA, we construct a timeline of genetic diversity, offering rare insight into how populations have changed over time. A case study on the threatened Charadriiform species, *Numenius Arquata* (Eurasian curlew), revealed increased inbreeding in the Bavarian population as well as a reduction in climatically suitable spaces for future projections of the species in this region, underscoring the value of this approach. By combining data from past and present populations, ecological niche modeling, and genomic analyses, we aim to (i) estimate ancestral population size and point of decline (ii) compare past and present models to determine if the change in suitable habitat for the species coincides with extinct populations and (iii) use genetic analyses to compare populations within regions which the model has deemed highly favorable and less favorable. This interdisciplinary approach - integrating demographic history reconstruction, population genomic analyses, and ecological niche modeling - provides a comprehensive framework for assessing extinction risk and guiding conservation efforts.

Sequence composition and conservation predict the phenotypic relevance of transposable elements

Yari Cerruti^{1,2}, Daniela Fusco², Mahdis Jahanbin², Davide Marnetto², Paolo Provero²

¹School of Mathematics and Statistics, University of St Andrews, St Andrews, United Kingdom

²Department of Neurosciences “Riva Levi Montalcini”, University of Turin, Turin, Italy

Transposable elements (TEs) are powerful drivers of genome evolution, in part through their ability to rapidly rewire the host regulatory network by creating transcription factor binding sites that can potentially be turned to the host's advantage in a process called exaptation. However, the effects on the host phenotype vary widely among different TEs. In this study, we classify TEs based on their contribution to the human host phenotypes at both molecular and macroscopic scales. TE contributions to chromatin accessibility, gene expression, and the heritability of complex traits are strongly correlated to each other, confirming that the main mechanism through which TEs affect the host phenotype is through the rewiring of the regulatory network. TE sequence and evolutionary features are able to explain a large fraction of the variance in phenotypic relevance, and in particular more than 50% of the variance of their contribution to the heritability of complex traits. A conspicuous exception to this pattern is represented by TEs of the ERV1 family, whose phenotypic impact cannot be explained by our model: In particular, this family includes a set of relatively young TEs whose phenotypic relevance is much larger than would be expected based on their sequence and evolutionary parameters. These TEs are involved in fast-evolving biological processes related to the interaction of the organism with its environment. In conclusion, our results confirm quantitatively that TE insertions affect the host phenotype mostly through the rewiring of its regulatory network; identify a signature of phenotypic relevance based on sequence and conservation properties; and highlight several TEs as promising candidates for functional studies.

Species delineation using genome-wide D_{XY} - and F_{ST} -estimates

Menno J. de Jong

Senckenberg Biodiversity and Climate Research Centre (BiK-F), Frankfurt am Main, Germany

Accurate and consistent species delineation is essential for effective nature conservation. We performed a meta-analysis of genomic data of mammalian sister lineages to evaluate whether species boundaries can be identified using estimates of genome-wide sequence dissimilarity. We find that combined criteria of absolute (D_{XY}) and relative (F_{ST}) genetic distance can predict IUCN taxonomic status with high accuracy. This predictive power demonstrates the feasibility of the genetic species concept, according to which a pair of populations may be diagnosed as separate species if their genetic distance is within the range typically observed for sister species. Lineage pairs with sterile hybrid offspring have genetic distances well above species boundaries, indicating that the celebrated biological species concept is rarely implemented in practice. Our findings further question the relevance of bottleneck speciation and hybrid speciation, and instead are more consistent with parapatric speciation. In conclusion, we suggest a simple and universal set of objective rules for species delineation, and present new evidence in favour of a particular speciation mode.

The genetic bases of quantitative morphological variation in a population of *Drosophila*

İdil Çağla Ertaş, Nicolas Gompel

Bonn Institute for Organismic Biology, University of Bonn, Bonn, Germany

Morphological differences between species may find their origin in standing genetic variation available in populations of their common ancestor. To conceptually link these two evolutionary timescales – populations and species– in terms of genetic mechanisms underlying morphological variation, we are comparing pigmentation variation on fly wings among closely related species, to pigmentation variation in a population of one of these species. The lab's previous work paints a fairly detailed picture of the genetic origin of inter-species differences in wing pigmentation, notably for a spot of dark pigmentation at the wing tip. I will present a new project aiming at deciphering the genetic origin of variation in wing pigmentation within a population of a single species with a wing spot, *Drosophila biarmipes*. I will first present a thorough quantitative characterization of the phenotypic variation in the subject population. I will then present our plans to identify the genetic and molecular bases of this variation and compare it to among-species genetic variation.

To map the genetic determinants of population-level variation, we will first carry out a classical high-resolution GWAS. We will functionally test the regions of interest with classical genetics using custom transgenics to interfere with the corresponding genes. Independently, we will particularly scrutinize the possible role of the gene *yellow* in pigmentation variation at the population level, as we recently found that regulatory changes at this gene explain continuous variation in wing pigmentation intensity among species.

Our project, at the interface of population genetics and Evo-Devo will result in the possibility to compare the genetic make-up of phenotypic changes among species to the genetic architecture of variation within species.

Genomic erosion footprint in insects under extreme decline and isolation

Elisabeth Karalashvili¹, Aryadevi Anitha Shaji¹, Nina Casillas¹, Astrid Böhne¹, Martin Husemann², Bruno Hüttel³, Peer Schnitter⁴, Volker Neumann⁵, Tobias Meitzel⁶, Lothar Buttstedt, Eckart Stolle¹

¹Leibniz Institute for the Analysis of Biodiversity Change, Museum Koenig Bonn, Bonn, Germany

²State Museum of Natural History Karlsruhe, Germany

³Max Plank Genome Center Cologne, Germany

⁴Landesamt für Umweltschutz Sachsen-Anhalt

⁵Institute of Biology, Martin-Luther-University Halle-Wittenberg, Germany

⁶Leibniz Institute of Plant Genetics and Crop Plant Research, Germany

Biodiversity is declining globally. Such rapid collapse may pose stark challenges for species and populations due to increases in inbreeding and genetic drift, decreases in genetic diversity and inefficacy of selection. Long-term population fragmentation may also lead to isolation and divergence between populations, and, consequently, potentially mirroring genomic effects observed in nascent species.

Understanding the genomic processes underlying population declines and extreme isolation is essential not only to determine genomic erosion in species and populations for conservation purposes, but also to compare genomic effects to those naturally occurring during bottlenecks and nascent speciation. Most research focused on vertebrate species, largely overlooking the vast amount of biodiversity: insects and other invertebrates. A high diversity of different life histories gives rise to the expectation that species and populations are impacted or react differently to decline or additional factors such as historical (anthropogenic) influences, bottlenecks, historically small population sizes or long-term geographic isolation. Understanding the genomic dynamics and effects in non-model species is crucial for developing more effective conservation strategies as well as a better understanding of the species diversity we observe in the world.

Here we take advantage of advancements of sequencing technologies, including museomics, to start to gain knowledge of the genomic footprint of biodiversity decline in non-model or non-vertebrate species, including two beetles, a bush cricket, a butterfly, a damselfly and a bee, representing the large insect orders. These species are examples of extreme decline and/or extreme isolation. Population genomics and museomics paired with novel genome assemblies will enable us to uncover the extreme-case genomic signatures of biodiversity decline. This will improve our knowledge for large parts of biodiversity.

Lecture: Polymorphism-aware phylogenetic models (PoMos) and Selection

Carolin Kosiol¹, Rui Borges²

¹Centre for Biological Diversity, University of St Andrews, St Andrews, United Kingdom

²School of Mathematics and Statistics, University of St Andrews, St Andrews, United Kingdom

The availability of population genomic data through new sequencing technologies gives unprecedented opportunities for estimating important evolutionary forces. Yet, analytical methods that can handle polymorphisms jointly with sequence divergence across species are rare and not easily accessible to empiricists.

We implemented polymorphism-aware phylogenetic models (PoMos), an alternative approach for species tree estimation, in the Bayesian phylogenetic software RevBayes. PoMos naturally account for incomplete lineage sorting, which is known to cause difficulties for phylogenetic inference in species radiations, and scale well with genome-wide data. Simultaneously, PoMos are designed to disentangle the interplay of mutation, genetic drift, and directional selection (GC-biased gene conversion), along with the previously unexplored balancing selection pressures on ultra-long timescales comparable with species divergence times by analysing multi-individual genomic and phylogenetic divergence data.

Workshop: PoMoBalance and RevBayes

Carolin Kosiol

Centre for Biological Diversity, University of St Andrews, St Andrews, United Kingdom

We implemented polymorphism-aware phylogenetic models (PoMos), an alternative approach for species tree estimation, in the Bayesian phylogenetic software RevBayes. PoMos naturally account for incomplete lineage sorting, which is known to cause difficulties for phylogenetic inference in species radiations, and scale well with genome-wide data. Simultaneously, PoMos are designed to disentangle the interplay of mutation, genetic drift, and directional selection (GC-biased gene conversion), along with the previously unexplored balancing selection pressures on ultra-long timescales comparable with species divergence times by analysing multi-individual genomic and phylogenetic divergence data.

In the workshop we will do a hands-on exercises using RevBayes and small primate data sets.

PoMo-cod: a polymorphism-aware phylogenetic codon model

Ioanna Kotari^{1,2}, Carolin Kosiol³, Rui Borges⁴

¹Institut für Populationsgenetik, University of Veterinary Medicine, Vienna, Austria

²Vienna Graduate School of Population Genetics, Vienna, Austria

³Centre for Biological Diversity, University of St Andrews, St Andrews, United Kingdom

⁴School of Mathematics and Statistics, University of St Andrews, St Andrews, United Kingdom

Modelling sequence evolution offers crucial insights into the forces shaping genetic diversity. DNA models capture how mutations introduce variation, while amino acid models reveal how natural selection drives evolutionary change. Coding sequences are especially informative, as they reflect both mutational input and selective pressures at the protein level. Existing codon models provide a framework for testing adaptive evolution by incorporating the nonsynonymous to synonymous substitution rate ratio (dN/dS), which allows us to discern the direction and magnitude of natural selection. However, current models often lack biological realism by neglecting factors such as GC-biased gene conversion (gBGC), selection on synonymous codons, and within-species polymorphisms, all of which can bias dN/dS estimates and produce false positives. In response, we present a new model, PoMo-cod (Polymorphism-aware Phylogenetic Model for Codon Evolution), an extension of existing polymorphism-aware DNA models adapted to the codon level. PoMo-cod advances codon modelling by integrating mutation, gBGC, and selection at the population level, using polymorphic sites to more accurately describe sequence evolution. We apply this framework to explore how the interaction of these adaptive and non-adaptive processes affect genome-wide scans for positive selection.

Genetic structure, spectral characteristics and evolutionary history of butterflies in the European *Polyommatus eros* species group (Lepidoptera, Lycaenidae)

Nok Ting Lam¹, Vladislav Ivanov^{1,2}, László Péter Biró³, Zsolt Bálint^{3,4}, Krisztián Kertész³, Gábor Piszter³, Leonardo Dapporto⁵, Roger Vila⁶, Marko Mutanen¹, Vlad Dincă^{1,7}

¹ Ecology and Genetics Research Unit, University of Oulu, Oulu, Finland

² Department of Biogeography, Faculty VI, University of Trier, Trier, Germany

³ Institute of Technical Physics and Materials Science, HUN-REN Centre for Energy Research, Budapest, Hungary

⁴ Hungarian Natural History Museum, Budapest, Hungary

⁵ ZEN lab, Dipartimento di Biologia, University of Florence, Sesto Fiorentino, Italy

⁶ Institut de Biologia Evolutiva (CSIC-Universitat Pompeu Fabra), Barcelona, Spain

⁷ “Grigore Antipa” National Museum of Natural History, Bucharest, Romania

Species complexes with allopatric populations scattered across a wide geographic area represent some of the most challenging topics for taxonomy, but also for the prioritization of limited conservation resources. A prime example is represented by cold-adapted butterflies of the *Polyommatus eros* species group, with a fragmented distribution across Eurasia. In Europe, this group was often regarded as comprising two species (*Polyommatus eros* (Ochsenheimer) and *Polyommatus eroides* (Frivaldszky)), although different taxonomic views recognized from one to four species. Based on limited material and genetic data, a single species (*P. eros*) is recognized by the most recent checklists of European butterflies.

Here we use genome-wide data (double digest RAD sequencing) coupled with information on wing spectral characteristics to investigate evolutionary relationships among European butterflies of the *P. eros* species group. Taken together, results of phylogenetic analyses, genetic structure, gene flow, species delimitation and wing reflectance revealed a complex pattern likely shaped by the Pleistocene glaciations. While differentiation between some populations is relatively low, two lineages (one detected in mountains of northern Albania and Montenegro, the other, assignable to taxon *menelaos*, endemic to southern Greece), appear to be undergoing accelerated differentiation, suggesting a process of paraphyletic speciation. The endangered taxon *menelaos* is particularly differentiated in terms of genetics and wing spectral properties, the latter suggesting reinforcement with respect to Balkan *eroides*. Although assessed at a continental scale, a more complete understanding of the intricate patterns displayed by this species group will only be possible with the inclusion of extra-European populations.

Evolutionary analyses of antibiotic resistance in *Acinetobacter baumannii*

Roni Louden

Centre for Biological Diversity, University of St Andrews, St Andrews, United Kingdom

ATP phosphoribosyl transferase (ATPPRT) is an enzyme in *Acinetobacter baumannii*, a bacterium known for posing an overwhelming threat to the population primarily due to its antibiotic resistance, that is essential for its survival in the lungs. The use of computational biology will elucidate the evolutionary history of ATP phosphoribosyl transferase and analyse variability between strains to deepen our understanding of the enzyme. Specifically of interest is the short and long form of catalytic domain HisG that is found in nature and the evolutionary order of these genes.

Lecture: Neural networks

Christoph Mayer, Peter Arsic

Leibniz Institute for the Analysis of Biodiversity Change, Museum Koenig Bonn, Bonn, Germany

Machine learning is becoming a standard tool in many fields of biology. In this lecture we will introduce neural networks as well as often used neural network architectures. We will start with networks consisting of a series of fully connected layers used for classification tasks. Then we will introduce convolutional layers for feature extraction in images, sequences and multiple sequence alignments. Finally, the U-Net, used for image segmentation will be introduced.

Workshop: Machine learning topics in sequence analyses

Christoph Mayer, Petar Arsic

Leibniz Institute for the Analysis of Biodiversity Change, Museum Koenig Bonn, Bonn, Germany

Convolutional neural networks can not only be used for image analyses, such as object detections, segmentation and classification tasks, but also when working with molecular sequences. This workshop will demonstrate how to encode molecular sequences for use in neural networks. Then we will demonstrate the usage of neural networks in two projects in which we train neural networks for (i) topology predictions on four sequence alignments, (ii) conducting multiple sequence alignments. After neural networks have been trained and saved, they can be used for prediction tasks on previously unseen data.

Workshop: DeepBLUES: Deep Learning for Balancing, Linkage, and Effects of Selection

Antonio Pacheco, Nick Bailey

Centre for Biological Diversity, University of St Andrews, St Andrews, United Kingdom

Balancing selection (BS) leaves distinct signatures in the genome that can be challenging to distinguish from patterns generated by phenomena like linkage disequilibrium (LD) or demographic processes (e.g. changes in population size). In this workshop, we will develop a deep learning framework to classify genomic regions under BS versus those not under BS, which is to say neutrality. We will use genomic data simulated under an Out-of-Africa human demographic model generated by stdpopsim, which has been varied to simulate BS and neutrality. These simulations generated both variant call data and ancestral recombination graphs (ARGs). We will compute population genetic summary statistics (e.g. those relating to site-frequency spectra) from the variant data, while phylogenetic statistics (e.g. those relating to tree balance) will be computed from individual trees in the ARGs. These data will be used as input for a deep learning model to classify the different scenarios.

Lecture: PhyloCNN

Manolo F. Perez

Imperial College London, London, United Kingdom

PhyloCNN is a simulation-based deep learning method that encodes phylogenetic trees using local node features and applies a dedicated convolutional architecture for parameter estimation and model comparison in birth–death models. By varying the neighbourhood context used to describe each node’s local features, we show that broader context improves accuracy—particularly on smaller training sets and for complex models. Against other recent deep-learning approaches, PhyloCNN matches or exceeds accuracy across all parameters despite using one to two orders of magnitude fewer training simulations. Applied to real-world datasets on HIV superspreading in Zurich and primate seed-disperser diversification, PhyloCNN demonstrates high accuracy and computational efficiency, enabling analyses tailored to specific phylogenetic histories and sampling scenarios.

Uncovering the architecture of selection from experimentally evolved populations with Machine Learning

Diogo Ribeiro¹, Rui Borges¹, Christian Schlötterer²

¹School of Mathematics and Statistics, University of St Andrews, St Andrews, UK

²Institute of Population Genetics, University of Veterinary Medicine Vienna, Austria

Experimental evolution enables us to study how populations adapt to new environments over several generations. Combined with high-throughput sequencing techniques, time-series genomic data promises to answer how organisms respond and evolve in the face of controlled environmental challenges. The changes in allele frequency observed in these datasets result from the interaction of various evolutionary forces, such as genetic drift, selection, and recombination. The challenge lies in capturing adaptive forces, particularly in polygenic scenarios where many loci contribute, possibly leaving similar traces. Machine Learning (ML) has the potential to characterize the genetic architecture of adaptation, providing a promising alternative to classic population genetic tests. Here, we employed ML techniques to predict the adaptive architecture of a *Drosophila melanogaster* experimental population founded by two inbred strains and undergoing adaptation to hot temperatures for 210 generations. Specifically, we aim to infer the effect size of a given genomic region, which captures the combined effect of the number of selection targets and their selection strength. We trained multiple models using simulated time-series data, replicating the experimental design. The models were employed to analyze entire chromosomes, capturing global patterns, and using a moving window approach for finer resolution. We observed that the accuracy of the inferred effect sizes decreased with reduced window size, a trend attributed to stronger linkage in smaller windows. Our study provides valuable insights into the number of players and their relative contribution to the polygenic response to adaptation.

Lecture: Does experimental evolution recapitulate speciation genomics?

Mike Ritchie

Centre for Biological Diversity, University of St Andrews, St Andrews, United Kingdom

“Evolve and Resequence” studies are becoming a common way of testing evolutionary theory, and allow detailed tracking of changes in gene frequencies under short to medium time scales. Do results repeat patterns seen comparing species and give extra insight into patterns of divergence? I will describe and contrast two experimental evolution studies; a long-term one examining divergence under sexual selection and a short-term one looking for evidence of adaptive introgression between two species with strong post-mating barriers.

Adding Morphological Traits to the Species Tree inference under a Polymorphism-Aware Phylogenetic Models approach

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In evolutionary biology, genomic sequences and morphological data play a crucial role in the study of evolutionary histories. However, for species tree inference and species delimitation these kinds of data have been treated as different datasets for analytical purposes, weighting one over the other or just analysing them sequentially without a true relationship, which in some scenarios could lead to different or contradictory results. I intend with this project to develop a species delimitation algorithm that combines both continuous morphological traits using Brownian motion and Ornstein-Uhlenbeck processes and multi-locus genomic data using Polymorphism-aware Phylogenetic Models (PoMos) into a common model that explicitly considers both types of data. For the implementation I am using the open-source probabilistic language RevBayes, which offers several advantages for phylogenetic modelling and validation. In order to test the model, multiple datasets will be generated using the traits and genes simulator implemented as part of the iBPP software.

Furthermore, I am going to test the model using real morphological data and sequences from museum samples of South American Didelphidae species and compare with other species delimitation algorithms; to improve the current knowledge we have of the evolutionary history of this clade.

Evolutionary dynamics of GRCs in passerine birds

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Understanding how genomic elements contribute to lineage divergence across evolutionary time is key to speciation. The Germline-Restricted Chromosomes (GRCs) in Passerine birds offer a unique system to explore this, as they evolve exclusively in the germline and are eliminated from somatic cells. GRCs has been well documented in songbirds (oscines), yet its broader evolutionary significance across passerines remains unclear. Previous studies have been focused on species closely related to the zebra finch, leaving other major passerine lineages, the suboscines, unstudied. Here, we present the most extensive comparative analysis of GRCs to date, spanning ~50 million years of passerine evolution. Using three high-quality germline reference genomes and 25 additional germline genome drafts from all major passerine clades, we show that GRCs are present in approximately two-thirds of all surveyed species. These chromosomes exhibit rapid and independent evolutionary trajectories, diverging markedly from the conserved structure of A chromosomes. Despite their dynamic nature, GRCs retain a subset of genes under purifying selection, including deeply conserved elements such as *cpeb1* and *pim1*, as well as lineage-specific recurrent acquisitions like *bmp15*, which has independently entered the GRC up to six times. We also report the presence of *zglp1* in an ancestral GRC, following its duplication from chromosome 30 and loss from the core genome, suggesting a role for the GRC in preserving essential functions. This work establishes a genomic and evolutionary framework for studying GRCs as non-Mendelian, potentially selfish elements. Their broad phylogenetic distribution and gene content dynamics open new opportunities to study genomic conflict across lineages and evolutionary timescales.

Lecture: The contribution of genomic (in)stability to population and species divergence

Alexander Suh

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Disentangling the causes and consequences of the divergence between populations and emerging species requires an understanding of the mutational forces at play. Birds have long been regarded as a puzzling case where, despite having the largest number of extant amniote species, chromosomes are highly syntenic and collinear, and genes are rarely lost or duplicated. Here I review recent developments suggesting a "two-speed" genome of birds through the hidden depths of genetic diversity (e.g., centromeres, sex chromosomes, and germline-restricted chromosomes) and how such unstable parts of avian genomes may affect population and species divergence.

A Machine Learning approach to identifying recent selection in blue whale evolution

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The effect of recent human activity on whales has had a well-documented negative impact on their populations. For the blue whale (*Balaenoptera musculus*), the largest of the cetaceans, industrial whaling led to at least ten-fold population decline between 1900 and 1970, the Antarctic population was the most affected. However, little is known about the specific genomic changes that occurred between the pre-whaling and post-whaling periods. Proposed adaptive phenotypic shifts, such as a decrease in body size, suggest that selection may have acted on certain traits during this time, but the genetic basis remains largely unexplored. Here, we are developing a computational workflow to identify potential selective sweeps by comparing the genome of a newly sequenced Antarctic blue whale individual from around 200 years ago—representing the pre-whaling era—with genomes from modern blue whales. The analysis will focus on local patterns of heterozygosity and allele frequency change over time. By identifying regions where modern genomes show reduced genetic diversity relative to the ancient sample, we aim to detect signals of recent selection and demographic change. These genomic features will be used to train a supervised machine learning methods (ensemble approach that combines Random Forest and Gradient Boosting), informed by simulations under both neutral and selective scenarios. The goal is to build a predictive framework for identifying candidate regions shaped by recent evolutionary pressures. This will allow to improve our understanding of how large-scale human activity may have influenced the evolutionary trajectory of whales.

St Andrews stuff

Random thoughts for a short visit by Mike Ritchie.

(Official tourist advice: <https://www.visitscotland.com/places-to-go/st-andrews>)

Watering holes

There are plenty of places to get a drink and nibble in St Andrews, with a nice range of beer and whiskies. For example:

St Andrews brewing company (typically called "brewco").

<https://www.standrewsbrewingcompany.com/venues/st-andrews-south-street>

Central bar

<https://www.belhaven.co.uk/pubs/fife/central-bar>

The Criterion (pricey)

<https://www.criterionstandrews.co.uk/>

The Whey Pat

<https://www.belhaven.co.uk/pubs/fife/whey-pat-tavern>

More generic hostelrys are the Brewdog, The Rule, The Saint

Places to eat

Fish & chips are essential. The two best places are *Tailend* and *Cromar's* on Market Street.

People argue about which is best, but it's Tailend.

<https://www.thetailend.co.uk/>

<https://cromars.co.uk/>

Very good:

Tanon

Excellent Thai 'pop-up'.

Maisha

Indian, extensive menu and the seafood is good

<https://www.maisharestaurant.co.uk/>

The Adamson

<https://www.theadamson.com/>

Rogue (pricey)

<https://www.rogue-standrews.com/>

Little Italy

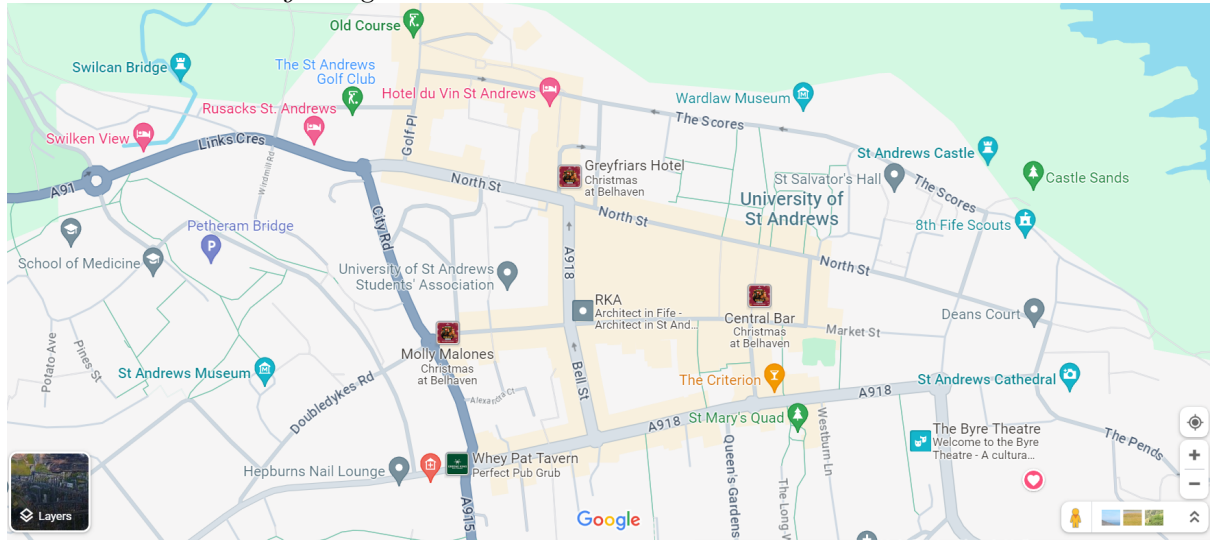
<https://littleitaly.has.restaurant/>

St Andrews also has lots of more standard restaurants and cafes.

Finally, do not miss a legend of Tayside; the best place for traditional pies, sausage rolls, fudge donuts, shortbread, amazing cakes and bread:

<https://www.fisheranddonaldson.com/our-shops>

A short walk to see some of the sights.



Starting from the North Haugh, walk towards the famous Old Course, strikingly familiar to anyone who has ever watched golf on the TV. The golf museum is fun for aficionados.



North of the golf is the 'West Sands', or the Chariots of Fire beach, perfect for a windy walk with lots of fresh air.

Heading east from the golf is a walk along the shore, past grand old houses (including the building in which Chambers wrote 'Vestiges of Creation' and the Principal's mansion), the University museum, then on to the Castle, with a visitor centre, dungeons, mines and counter-mines. And another small beach with an outdoor swimming pool, ideal if anyone fancies a swim.



Onwards takes you past the Cathedral then down to the harbour area. The view back to the castle and across to Angus is particularly fine. The harbour wall is a nice walk. It has just been reopened after it was damaged in a storm.





Onwards to East Sands and past the Scottish Oceans Institute (one of the three Centres that form the bulk of Biology here). Just before SOI is a grilled cheese shack that PhD student claim is the best in the UK.

Beyond East sands is a rough shoreline that contains numerous features, including Maiden Rock and some remarkable fossil trails.

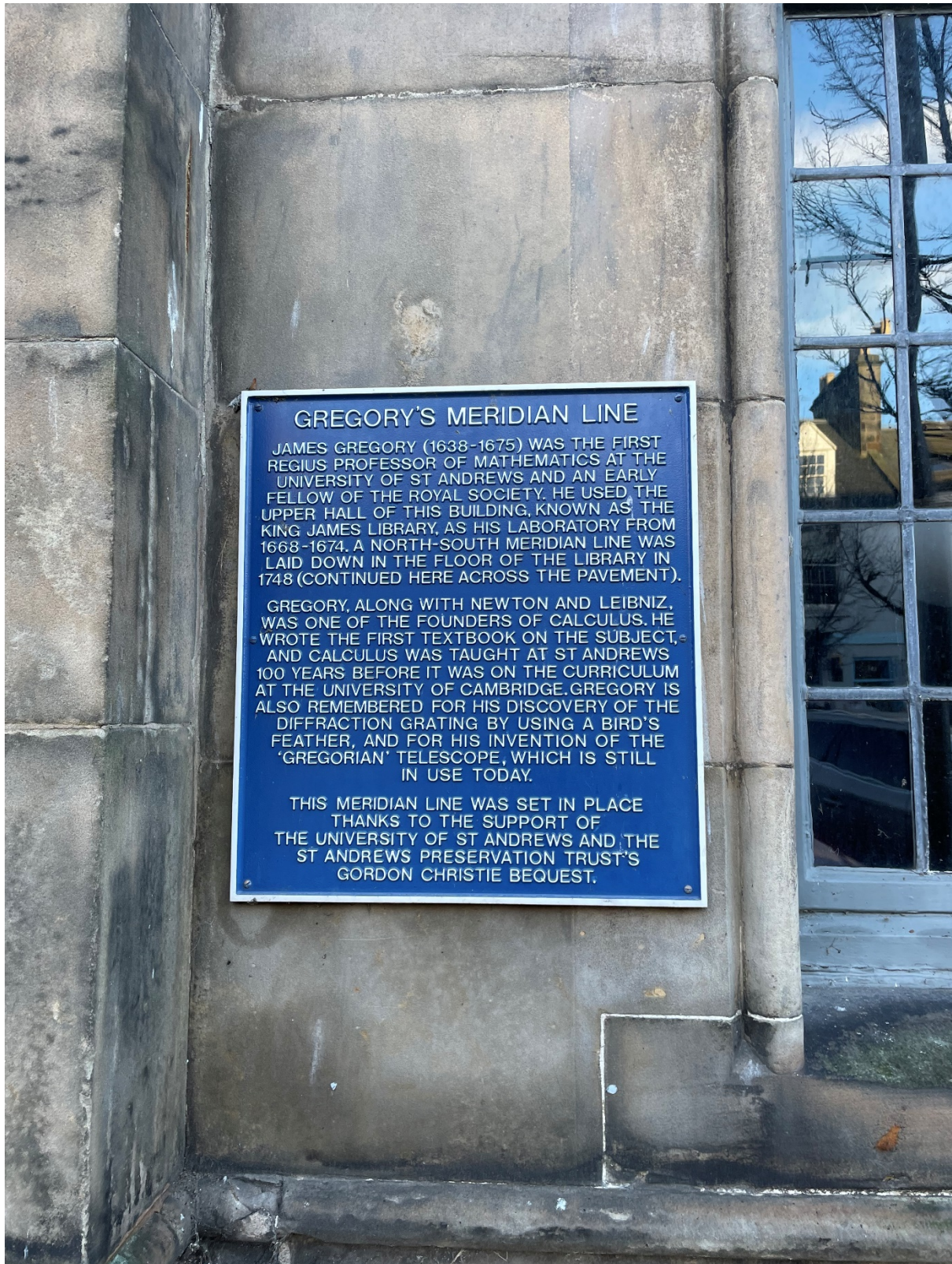
Heading back into St Andrews, on North Street is D'Arcy Thompson's house where he lived when working here. Pop into lovely St Mary's quad with the old Parliament Hall, where the old Scottish parliament sat when Edinburgh was plague bound. On the street outside is Gregory's meridian line, a forerunner to GMT (if precedence applied, we should be on St

Andrews Mean Time, not GMT). The Bell Pettigrew museum is worth a visit if open at this time of year, and includes more D'Arcy Thompson memorabilia and a thylacine.









On South Street is the centre of the University containing College Gait, St A Salvator's and College hall. Anyone who watched 'The Crown' will recognise this bit of the University.